Arbeitsauftrag Freitag 06.03.2026

Dokumentation

1. & 3.

```
PS C:\Users\Administrator> Get-DnsServerResourceRecord 3bhits.local

HostName                  RecordType Type       Timestamp            TimeToLive      RecordData
--------                  ---------- ----       ---------            ----------      ----------
@                         NS         2          0                    01:00:00        win-
eerp8aq2itt.
@                         SOA        6          0                    01:00:00        [3][win-
eerp8aq2itt.][hostmaster.]
3bhits.local              NS         2          0                    01:00:00        win-
eerp8aq2itt.
3bhits.local              SOA        6          0                    01:00:00        [3][win-
eerp8aq2itt.][hostmaster.]
ns1.3bhits.local.3bhit... A          1          0                    01:00:00        10.0.6.10
www.3bhits.local          A          1          0                    01:00:00        10.0.6.12
www                       A          1          0                    01:00:00        10.0.6.12


PS C:\Users\Administrator>
```

```
PS C:\Users\Administrator> Get-DhcpServerv4OptionValue -ScopeId 10.0.6.0

OptionId   Name            Type       Value                VendorClass     UserClass
PolicyName
--------   ----            ----       -----                -----------     ---------       -------
---
51         Lease           DWord      {691200}
3          Router          IPv4Add... {10.0.6.2}
6          DNS-Server      IPv4Add... {8.8.8.8}


PS C:\Users\Administrator>
```

2.

```
clie@clie:~$ ping 10.0.6.12
PING 10.0.6.12 (10.0.6.12) 56(84) bytes of data.
64 bytes from 10.0.6.12: icmp_seq=1 ttl=64 time=0.982 ms
64 bytes from 10.0.6.12: icmp_seq=2 ttl=64 time=0.393 ms
64 bytes from 10.0.6.12: icmp_seq=3 ttl=64 time=0.497 ms
64 bytes from 10.0.6.12: icmp_seq=4 ttl=64 time=0.473 ms
^C
--- 10.0.6.12 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3022ms
rtt min/avg/max/mdev = 0.393/0.586/0.982/0.231 ms
clie@clie:~$ ping www.3bhits.local
PING www.3bhits.local (10.0.6.12) 56(84) bytes of data.
64 bytes from 10.0.6.12: icmp_seq=1 ttl=64 time=0.285 ms
64 bytes from 10.0.6.12: icmp_seq=2 ttl=64 time=0.637 ms
64 bytes from 10.0.6.12: icmp_seq=3 ttl=64 time=0.493 ms
64 bytes from 10.0.6.12: icmp_seq=4 ttl=64 time=0.510 ms
^C
--- www.3bhits.local ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3044ms
rtt min/avg/max/mdev = 0.285/0.481/0.637/0.126 ms
clie@clie:~$ ping 10.0.6.2
PING 10.0.6.2 (10.0.6.2) 56(84) bytes of data.
64 bytes from 10.0.6.2: icmp_seq=1 ttl=128 time=5.55 ms
64 bytes from 10.0.6.2: icmp_seq=2 ttl=128 time=0.343 ms
64 bytes from 10.0.6.2: icmp_seq=3 ttl=128 time=0.312 ms
64 bytes from 10.0.6.2: icmp_seq=4 ttl=128 time=0.162 ms
^C
--- 10.0.6.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3027ms
rtt min/avg/max/mdev = 0.162/1.591/5.547/2.285 ms
clie@clie:~$
```

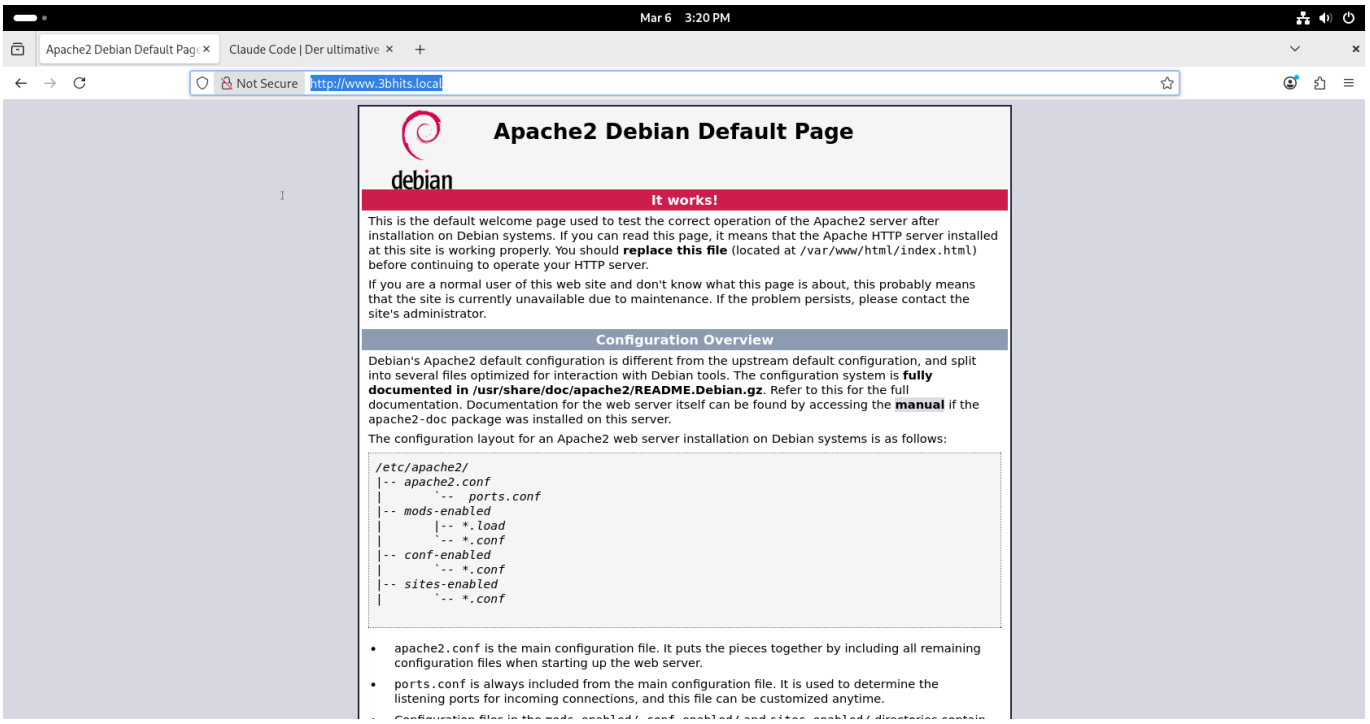
4.

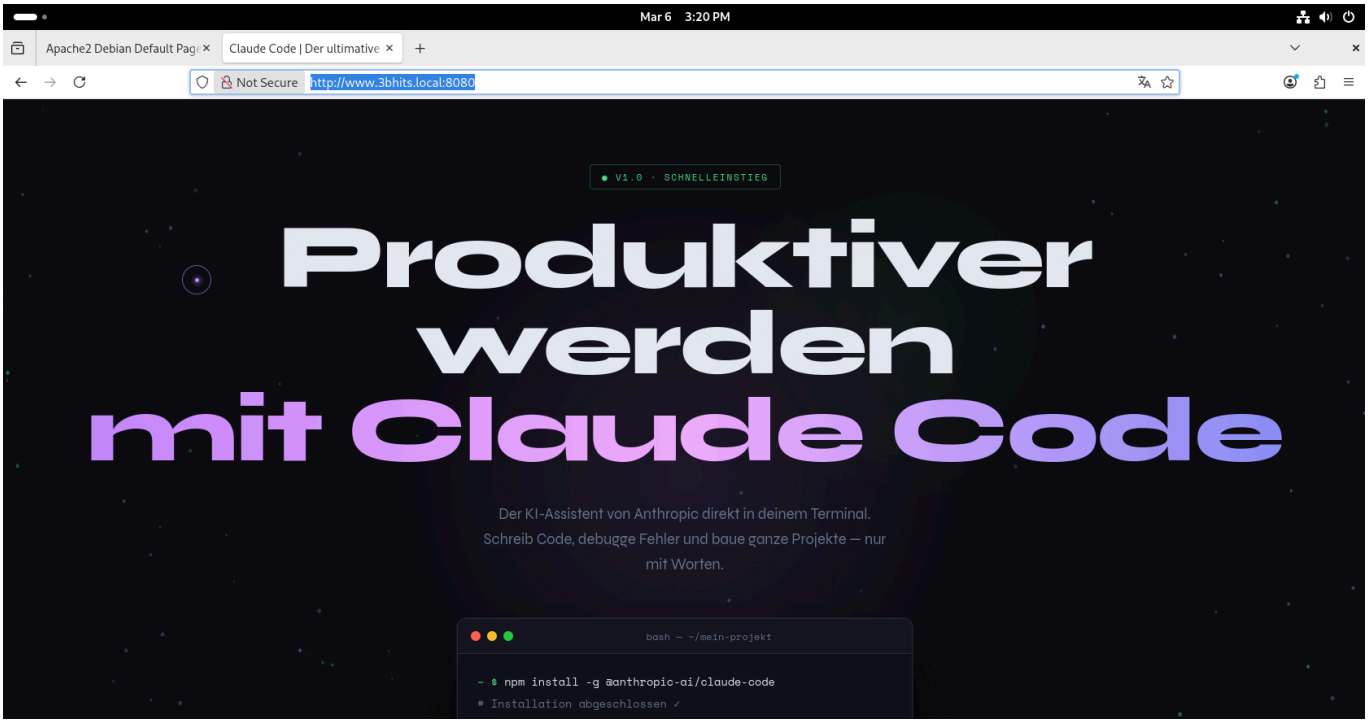
```
PS C:\Users\Administrator> Get-DhcpServerv4Reservation -Scopeid 10.0.6.10

IPAddress            ScopeId              ClientId             Name                 Type
      Description
---------            -------              --------             ----                 ----
      -----------
10.0.6.12            10.0.6.10            29-66-17-84-00-01... debweb               Both


PS C:\Users\Administrator>
```

5.





6.

DHCP

- Statische IPv4 Address

```
New-NetIPAddress -IPAddress 10.0.6.10 -InterfaceAlias "Ethernet0" -DefaultGateway 10.0.6.2 -AddressFamily ipv4 -Prefixlength 24
```

- ```
  Set-DnsClientServerAddress -InterfaceAlias "Ethernet0" -ServerAddresses "8.8.8.8", "8.8.4.4"
  ```
  ```

- **DHCP-Server installieren**

```
Install-WindowsFeature -Name DHCP -IncludeManagementTools
```

- Pool erstellen:

```
Add-DhcpServerv4Scope -Name frit -StartRange 10.0.6.55 -EndRange 10.0.6.155 -SubnetMask
255.255.255.0 -State Active
Get-DhcpServerv4Scope
```

- DNS-Server setzen `-Optionid 6`

```
Set-DhcpServerv4OptionValue -Scopeid 10.0.6.0 -Optionid 6 -Value "8.8.8.8"
Get-DhcpServerv4OptionValue -Scopeid 10.0.6.0
```

- Default-gateway setzen `-Optionid 3`

```
Set-DhcpServerv4OptionValue -Scopeid 10.0.6.0 -Optionid 3 -Value "10.0.6.2"
Get-DhcpServerv4OptionValue -Scopeid 10.0.6.0 # kontrollieren
```

- Reservations

```
Get-DhcpServerv4Lease -Scopeid 10.0.6.0
Add-DhcpServerv4Reservation -Scopeid 10.0.6.0 -Clientid (Get-DhcpServerv4Lease -Scopeid
10.0.6.0).Clientid -IPAddress 10.0.6.12 -Name "DWC" # Debien Webserver Client
```

DNS

- ```
  Install-WindowsFeature DNS -IncludeManagementTools
  ```

- ```
  Add-DnsServerPrimaryZone -Name "3bhits.local" -ZoneFile "3bhits.local"
  ```

- ```
  Set-DnsServerForwarder -IPAddress "8.8.4.4"
  Add-DnsServerForwarder -IPAddress "8.8.8.8"
  Get-DnsServerForwarder
  ```

- ```
  Add-DnsServerResourceRecordA -IPv4Address 10.0.6.10 -Name ns1.3bhits.local -ZoneName
  3bhits.local
  ```

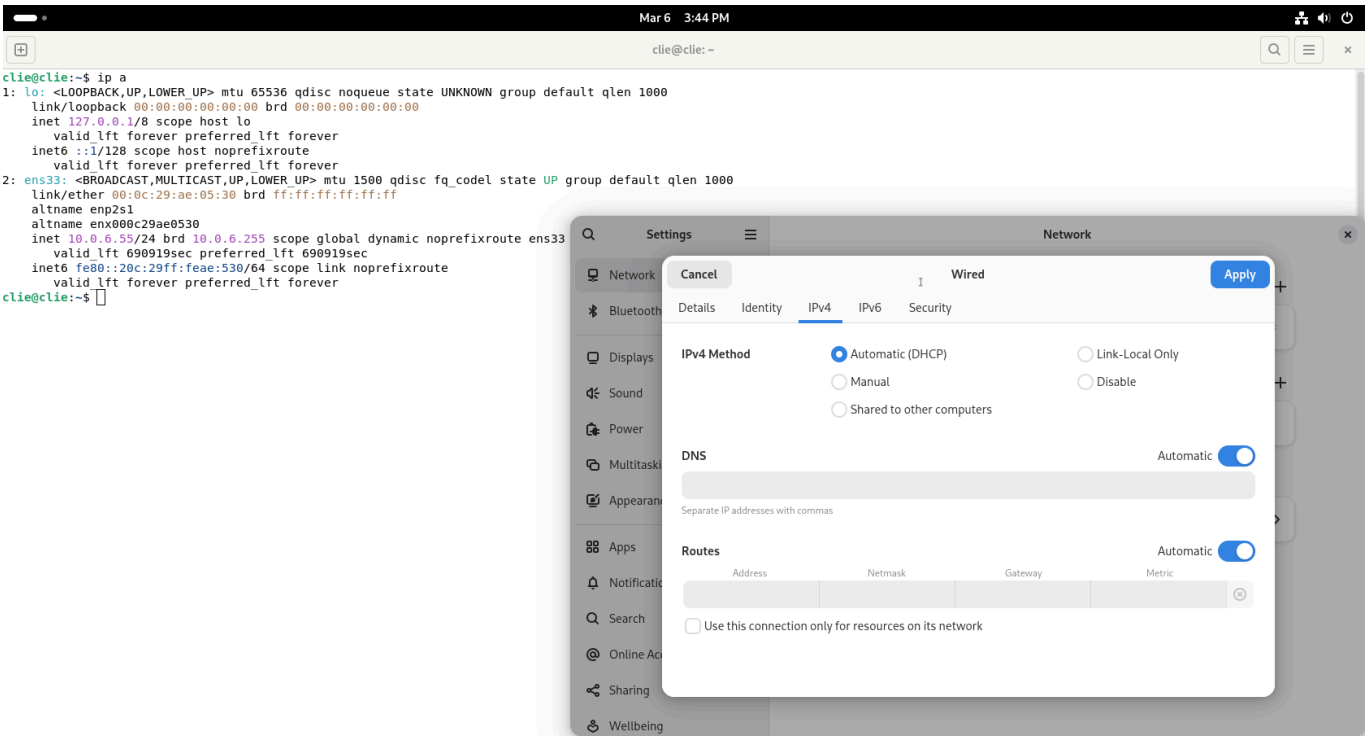
- ```
  Set-DnsClientServerAddress -InterfaceAlias Ethernet0 -ServerAddresses "10.0.6.10"
  ```

- ```
  Set-DhcpServerv4OptionValue -Scopeid 10.0.6.0 -DnsServer 10.0.6.10
  Get-DhcpServerv4OptionValue -Scopeid 10.0.6.0
  ```

- ```
  Get-DnsServerZone
  ```

- ```
  Get-DnsServerResourceRecord 3bhits.local
  ```

- ```
  Add-DnsServerResourceRecordA -ZoneName 3bhits.local -Name "www" -IPv4Address "10.0.6.12"
  ```

- ```
  Set-DhcpServerv4OptionValue -ScopeId 10.0.6.0 -OptionId 6 -Value 10.0.6.10
  ```

7.